

MotionGPT-2: A General-Purpose Motion-Language Model for Motion Generation and Understanding

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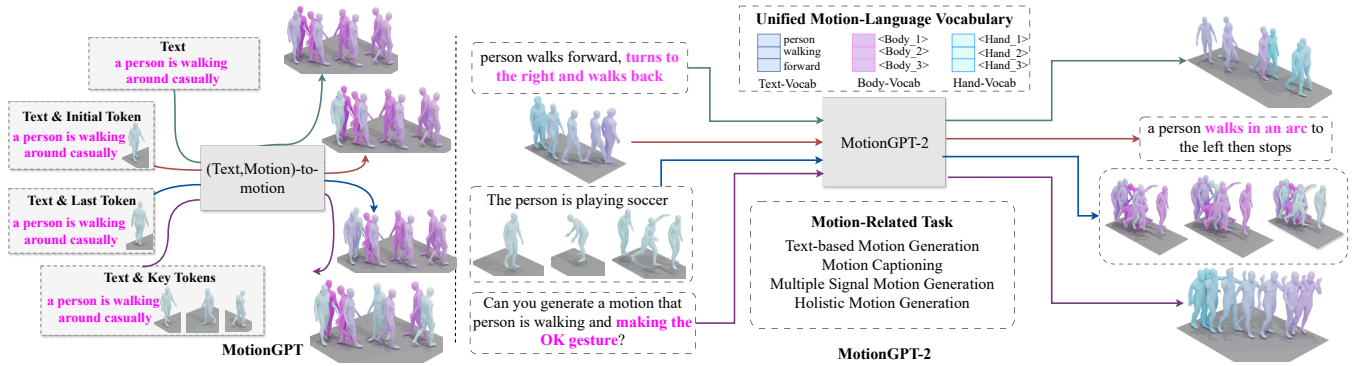


Fig. 1: This paper proposes a versatile motion-language framework via fine-tuned LLMs given different instructions, named MotionGPT-2. Compared with the previous MotionGPT [1], our MotionGPT-2 not only retains the unique capability of accommodating multiple control conditions, but also solve various motion-related tasks using a unified model.

Abstract—Generating lifelike human motions from descriptive texts has experienced remarkable research focus in recent years, propelled by the emerging requirements of digital humans. Despite impressive advances, existing approaches are often constrained by limited control modalities, task specificity, and focus solely on body motion representations. In this paper, we present MotionGPT-2, a unified Large Motion-Language Model (LMLM) that addresses these limitations. MotionGPT-2 accommodates multiple motion-relevant tasks and supports multimodal control conditions through pre-trained Large Language Models (LLMs). It quantizes multimodal inputs—such as text and single-frame poses—into discrete, LLM-interpretable tokens, seamlessly integrating them into the LLM’s vocabulary. These tokens are then organized into unified prompts, guiding the LLM to generate motion outputs through a pretraining-then-finetuning paradigm. We also show that the proposed MotionGPT-2 is highly adaptable to the challenging 3D holistic motion generation task. Enabled by the innovative motion discretization framework, Part-Aware VQVAE, which ensures fine-grained representations of body and hand movements. Extensive experiments and visualizations validate the effectiveness of our method, demonstrating the adaptability of MotionGPT-2 across motion generation, motion captioning, and generalized motion completion tasks.

Index Terms—3D Human Motion Generation, Large Language Models, SMPL, Vector Quantized-Variational AutoEncoder

I. INTRODUCTION

HUMAN motion [2]–[8] plays a pivotal role in various applications, including video gaming, robotic assistance, and VR/AR. Recent years have witnessed a rapid development of Generative Artificial Intelligence (GenAI) [9]–[15], paving the way for novel methods to motion synthesis.

Despite the impressive performance achieved by existing motion generation methods [16]–[19], current methods suffer from three major limitations: (1) *Limited control conditions*. Some text-to-motion approaches typically target only a single type of control condition, *i.e.*, either textual descriptions or multiple frame poses. This narrow scope constrains their practical applications in scenarios that require the generation of motion sequences conditioned on both text descriptions and multiple key-frame human poses, *e.g.*, social robotics and film animation. Although methods in [20]–[23] aim to integrate additional control modalities, they treat control signals as separate modalities, neglecting to construct a unified multi-modal representation. (2) *Task-specific frameworks without general world knowledge*. Existing models are often task-specific, such as diffusion-based and GPT-based frameworks [17], [24]–[26]. These methods lack the adaptability needed for multiple tasks (*e.g.*, captioning, in-betweening, prediction) and cannot fully leverage the world knowledge embedded in Large Language Models (LLMs). While recent efforts [27], [28] present unified frameworks, their reliance on relatively small language models means the broader potential of LLMs in motion understanding and generation remains unexplored. (3) *Body-only motion representations*. Existing text-based motion generation solutions [17], [24], [26], [29], [30] primarily focus on generating body-only motions rather than holistic motions. However, their plausibility and expressiveness remain unsatisfactory in certain scenarios (*e.g.*, sports activities and playing musical instruments), as important details of human motion—*e.g.*, fine-grained hand gestures, are

frequently overlooked.

To overcome these limitations, we propose MotionGPT-2, a unified Large Motion-Language Model (LMLM) capable of handling diverse control signals, performing various motion-relevant tasks, and generating holistic human motions. The key innovations of MotionGPT-2 include: (1) **Formulating Multimodal Control Signals into a Unified Representation**: MotionGPT-2 designs a versatile framework and task-aware prompts for human motion synthesis. This framework, in particular, allows for the generation of human motions governed by multimodal control conditions, described by $M_{\text{out}}, T_{\text{out}} = f(T_{\text{in}}, \text{task}, M_{\text{in}})$. Here M_{in} and T_{in} refer to the input motions and texts, while M_{out} and T_{out} represent the resulting outputs. The variable task indicates the task-aware prompts that adapt the model to specific motion-related tasks. Compared to MotionGPT [1], MotionGPT-2 broadens the scope by incorporating additional motion-related tasks, *i.e.*, motion captioning, general motion completion.

(2) **Building a Task-Agnostic Framework with Strong Generalization**: Our conference version, MotionGPT [1], argued that this challenge can be naturally addressed with the assistance of LLMs. There are several compelling reasons. First, recent investigations indicate that LLMs have the ability to comprehend inputs from multiple modalities (*e.g.*, images [31]–[34] and videos [35]) through lightweight adapters [36]. Therefore, we expect that with suitable adapters, LLMs will be capable of understanding motion sequences. Second, LLMs have learned a broad range of motion patterns from their extensive text training, which allows them to offer diverse human motion contexts for generating motion. Consequently, our motion generator, adapted from LLMs, can produce motions with a wide range of rich patterns. Third, because LLMs generate tokens in a sequential manner, generating human motion with adaptable sequences is now easily achievable.

In this work, we introduce MotionGPT-2, an enhanced version of MotionGPT [1] by using LLMs to jointly represent motion and language. We first embed the human motions into discrete motion tokens via the Vector Quantized Variational-AutoEncoder (VQ-VAE), which efficiently captures the underlying structure of human motion. Then, we expand the LLM’s vocabulary with these motion tokens, creating an enriched motion-language vocabulary. By incorporating human motion and language into a unified vocabulary, our model fosters a more transparent understanding of the complex relationships between the two. Further, MotionGPT-2 integrates tokens from both language and motion prompts to generate instructions. We implement a multimodal pre-training combined with an instruction-tuning approach to train MotionGPT-2 efficiently, utilizing the established LoRA adaptation method. The motion instruction tuning framework we have developed allows for the integration of pose sequence information into the fine-tuned LLM, while capitalizing on the strong motion priors inherent in the origin LLM. With mere 1% parameters, the generalized MotionGPT-2 achieves competitive results in multiple motion-related tasks compared to those trained-from-scratch models with specialized frameworks.

(3) **Achieving Precise Discrete Representations of Holistic Human Motions**: To address this issue, we incorporate

kinematic structure priors and design an innovative Part-Aware VQ-VAE for holistic motion tokenization. Compared to the vanilla motion VQ-VAE used in MotionGPT [1], the proposed Part-Aware VQ-VAE utilizes two-level discrete codebooks and motion encoders to learn body-hand representations. The key insight of our Part-Aware VQ-VAE lies in its ability to learn informative and compact representations of fine-grained holistic motions. This two-level discretization framework captures subtle hand movements while maintaining global body dynamics. Finally, the hand and body motion tokens are integrated into the LLM’s vocabulary, enabling MotionGPT-2 to correctly interpret and generate holistic motion-related sequences in response to instructions.

We conduct extensive experiments on the HumanML3D [30], KIT-ML [37] dataset, demonstrating that MotionGPT-2 has strong adaptability and efficiency across multiple motion-related tasks by fine-tuning an LLM. With only 1% of additional parameters, MotionGPT-2 achieves highly competitive performance across tasks while significantly reducing training time—requiring only 10% of the time compared to other methods. We also observe that joint training under multiple control instructions surpasses training with singular control signals, highlighting the effectiveness of the unified motion-language fine-tuning paradigm. Experiments on the Motion-X [38] benchmark verify that the MotionGPT-2 is adaptable to the challenging 3D holistic motion generation task. MotionGPT-2 sets a new benchmark for motion-related foundational models.

In summary, we extend our conference version [1] **by making several additional novel contributions**:

- We propose an enhanced version of the previous MotionGPT, MotionGPT-2. Compared to its predecessor, MotionGPT-2 serves as a unified Large Motion-Language Model to handle multiple motion-related tasks, enabling us to establish new state-of-the-art performance benchmarks.
- By creating an enriched motion-language vocabulary, we empower the pre-trained LLMs with the ability to unify the understanding and generation of body kinetics.
- We further extend the proposed MotionGPT-2 to tackle the challenging 3D whole-body motion generation task by introducing a whole-body motion discretization framework, **Part-Aware VQ-VAE**, which encodes body motions and hand gestures with two structured codebooks for fine-grained motion representations.

We conduct extensive experiments on the HumanML3D, KIT-ML, and Motion-X datasets to validate the superiority of our proposed LLM-based unified motion-language model across multiple motion-related tasks. We provide an in-depth analysis and visualizations of MotionGPT-2, indicating that further advancements in LLM technology hold the potential to enhance its performance in the future.

II. RELATED WORK

A. LLMs and Multi-modal Large Language Models

Large Language Models (LLMs) [39]–[44] have experienced rapid development recently, *e.g.*, BERT [39], GPT [40], and Google T5 [45]. Notably, models like GPT-4 [43] show outstanding performance on various language tasks due to

their extensive training datasets (GPT-4 utilizes 45 gigabytes) and numerous parameters. Traditionally, language models were crafted for specific tasks like translation or sentiment analysis, but recent innovations, such as ChatGPT, have broadened their functional scope. Built on the GPT-4 framework, ChatGPT can engage in interactive dialogues, showcasing strong natural language understanding. This effectiveness opens new avenues for downstream applications achievable through the fine-tuning of these LLMs. Nevertheless, fine-tuning these extensive models poses significant challenges. To mitigate this issue, several efficient fine-tuning techniques have emerged, including prompt tuning [46]–[48], adapters [49]–[51], and LoRA [36]. Our study is inspired by recent developments in LLMs while tackling a different problem through the integration of a novel modality.

B. Human Motion Generation

Motion generation has a long-standing history in research [16]–[18], [52]–[56] and can be conditioned on various inputs, including motion descriptions, specific actions, and music. For instance, HP-GAN [57] and [58] adopt a sequence-to-sequence model to forecast future poses from earlier ones. Moreover, ACTOR [59] utilizes a transformer-based VAE for both unconditional generation and action-driven motion generation. TRAJEVAE [60] generates motion sequences that align with a given trajectory when provided with an initial pose. Recently, a significant focus has been placed on text-conditional motion generation, which involves creating human motion sequences based on textual prompts. Previous researches [24], [30], [52], [61], [62] focus on modeling a joint latent space for motion and text alignment. In TEMOS [16], a VAE model is proposed that creates a shared latent space for motion and text interactions. T2M-GPT [24] formulates the motion generation as the next index prediction task and leverages small language models to model the translation mapping between discrete motion indices and text. MotionCLIP [52] and TM2T [63] align the shared space of text and motion with the expressive CLIP [64] embedding space. MotionDiffuse [17] introduces a diffusion model into its framework for generating motion from textual descriptions, which leads to impressive outcomes. In a different approach, MDM [18] aims to boost the coherence between motion and textual inputs by implementing CLIP [64] as the text encoder, thereby enhancing the model with more effective textual priors. However, it is challenging to model semantically complex relationships of human motions and texts, particularly in the absence of general world knowledge, *e.g.*, *how specific gestures convey intentions* and *how to interpret body language in different context*. Current efforts, such as MotionGPT [27], MotionLLM [65], M3-GPT [28] have initiated the development of a unified motion-language model aimed at generating plausible human motions along with textual descriptions driven by prompt instructions. Although remarkable achievements have been made in multiple motion-related tasks, these approaches, which only utilize relatively small language models, overlook exploration of the potential of LLMs in understanding and generating human motions. Moreover, they are inadequate in addressing the complex body-hand representations. Unlike prior approaches, MotionGPT-2 is distinguished as the first Large Motion-Language Model

(LMLM) capable of multimodal control, handling diverse motion-related tasks, and providing a comprehensive representation of motion.

III. MOTION GENERAL-PURPOSE GENERATOR (MOTIONGPT)

Our conference version MotionGPT acts as a Motion General-Purpose generator that is driven by multi-modal inputs. To align with the LLM’s token-in-token-out nature, we first quantize the human motions into code representations via the well-established motion VQ-VAE [66] (Section III-A). These motion discrete codes, along with text control conditions and specially-crafted task-aware instructions, are organized into a unified question template (Section III-B). By adjusting the task-aware instructions, MotionGPT reveals its potential as a generic baseline framework for tasks involving motion generation.

A. 3D Human Motion Quantization

VQ-VAE proposed in [66] is designed to learn discrete representations with semantic-rich codebooks in an encoding-decoding fashion. To discretely represent human motions as semantic tokens, we introduce the motion VQVAE model, which consists of a motion encoder $\mathcal{E}$, a motion decoder $\mathcal{D}$, and a codebook $\mathcal{B}_m = \{b_1, b_2, \dots, b_N\}$, where N denotes the codebook size. We denote a human motion as $\mathbf{m} = \{m_1, m_2, \dots, m_T\} \in \mathbb{R}^{T \times d}$, where T is the motion sequence length, and d is the dimension per frame.

To learn the codebook of human motions, the estimated motion embedding $\mathcal{E}(\mathbf{m})$ is transformed into a collection of codebook entries through quantization. Further, the quantized motion vector is obtained by searching the nearest corresponding embedding in a learnable codebook $\mathcal{B}_m$, which can be mathematically formulated as:

$$\mathbf{e} = \arg \min_{b_k \in \mathcal{B}} \|\mathcal{E}(\mathbf{m}) - b_k\|_2. \quad (1)$$

Based on the estimation latent representation $\mathbf{e}$, the motion decoder $\mathcal{D}$ employs several 1-D deconvolutional layers to extract frame-wise motion features, decoding the sequential codes $\mathbf{e}$ to the raw motion space as the motion $\hat{\mathbf{m}}$: $\hat{\mathbf{m}} = \mathcal{D}(\mathbf{e})$. The motion code p of the human pose $\mathbf{m}$ can be calculated as the index of its nearest embedding in the codebook, *i.e.*,

$$p = \arg \min_k \|\mathcal{E}(\mathbf{m}) - b_k\|_2. \quad (2)$$

The standard motion VQ-VAE can be trained by the three tailored loss: 1) a reconstruction loss to minimize the distance between the decoded motion $\hat{\mathbf{m}}$ and the origin motion $\mathbf{m}$, 2) a codebook loss to encourage the b_k to be drawn closer to the encoded embedding $\mathcal{E}(\mathbf{m})$, aligning the discrete code representation with origin motion embeddings. 3) a commitment loss to guide the encoded embedding $\mathbf{e}$ to remain close to the corresponding discrete code b_k , stabilizing the training process and reducing codebook oscillation. The total loss function $\mathcal{L}_{\text{VQVAE}}$ can be formulated as follows:

$$\mathcal{L}_{\text{VQVAE}} = \|\mathcal{D}(\mathcal{E}(\mathbf{m})) - \mathbf{m}\|^2 + \|\text{sg}[\mathcal{E}(\mathbf{m})] - \mathbf{e}\|_2^2 + \beta \|\mathcal{E}(\mathbf{m}) - \text{sg}[\mathbf{e}]\|_2^2. \quad (3)$$

Here, sg indicates the stop gradient operation and β is the hyper-parameter to control the weight of the commitment

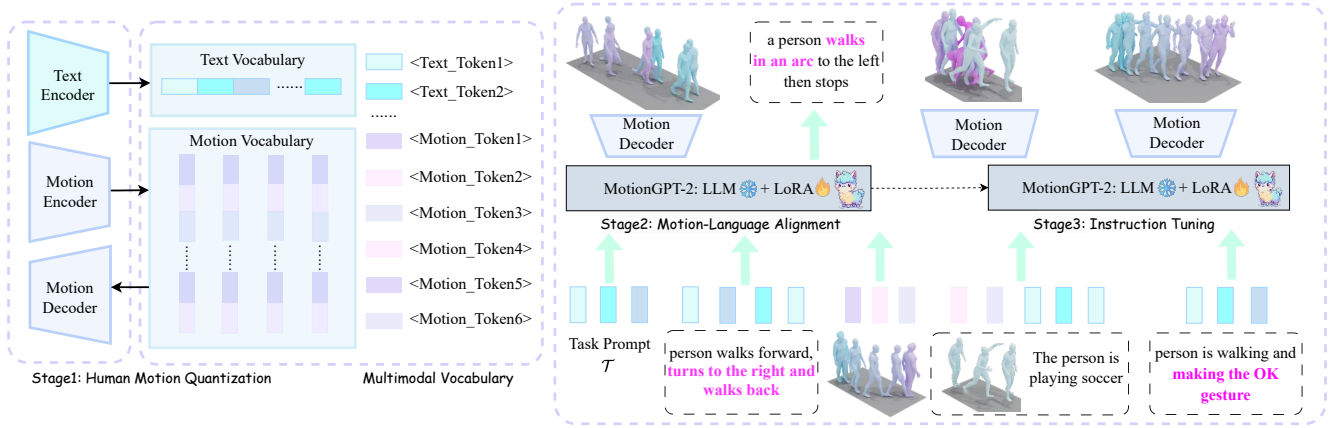


Fig. 2: MotionGPT-2 consists of multi-modal tokenizers (Section III-A) and a motion-language model (Section IV-B). With unified multimodal vocabulary and task-aware instructions (Section III-B). It enables to accept multiple control conditions (Section IV-C). MotionGPT-2 is learned by the *Motion-Language Alignment* stage and *Instruction Tuning* stage (Section IV-D).

loss. To enhance the quality of the generated motion, we incorporate *LI smooth loss* and *velocity regularization loss* in the reconstruction loss. The codebook is optimized with Exponential Moving Average (EMA) operation and codebook reset techniques following [1], [24], [27].

B. Instruction Generation

In our previous conference version, we design instructions that integrate task prompts and control conditions to enable (text, motion)-motion generation tasks. Given the task prompts $\mathcal{T} = \{t_1, t_2, \dots, t_{n_t}\}$, the text control conditions $\mathcal{X} = \{x_1, x_2, \dots, x_{n_x}\}$ and the pose control conditions $\mathcal{P} = \{p_1, p_2, \dots, p_{n_p}\}$ where n_t , n_x and n_p are the number of codes in $\mathcal{T}$, $\mathcal{X}$ and $\mathcal{P}$, the instruction template $\mathcal{I}$ is formulated as:

% General control conditions format

Control Conditions: {The Text control condition $\mathcal{X}$ $\langle x_1, x_2, \dots, x_{n_x} \rangle$ } {The Pose control condition $\mathcal{P}$ $\langle p_1, p_2, \dots, p_{n_p} \rangle$ }

% General instruction format

Instruction ($\mathcal{I}$): {The Task Prompt $\mathcal{T}$ $\langle t_1, t_2, \dots, t_{n_t} \rangle$ } {Control Conditions}

Pose control conditions $\mathcal{P} = \{p_1, p_2, \dots, p_{n_p}\}$, representing pose codes produced by the previously mentioned motion VQ-VAE. The entire instruction $\mathcal{I}$ is conceptualized as a series of specialized text inputs. By formulating diverse instruction prompts, the MotionGPT [1] tackles both traditional motion generation tasks and emerging challenges in motion synthesis.

Specifically, for text-based motion generation task, MotionGPT address it by instantiating following instruction $\mathcal{I}$:

Instruction ($\mathcal{I}$): {**Task Prompts:** “Generate a sequence of motion tokens matching the following human motion description.”} {**Control Conditions:** Text control condition $\mathcal{X}$ }

By adjusting instructions, MotionGPT is seamlessly adapted to a wide variety of control conditions, *e.g.* the textual description and an arbitrary number of human poses, making it a highly versatile and flexible solution for motion-related tasks.

C. Model Optimization

In the previous MotionGPT [1], we utilize a decoder-only LLM following [65], which effectively handles complex relationships and underlying patterns between text and motion. The token-in-token-out LLM maximizes the probability $p_\theta(x_t | x_{<t}, \mathcal{T}, c)$ of succeeding token in an autoregressive manner. Here, $\mathcal{T}$ represents the Task Prompts, c is the control signal, $x_{1:T}$ denotes the target token sequence. Therefore, during the training process, the Cross Entropy loss is applied to ensure the correspondence between the estimated tokens and the real tokens, fine-tuning the LLM by LoRA [36], which is mathematically formalized as:

$$\mathcal{L}_{\text{LoRA}} = - \sum \log p_\theta(x_t | x_{<t}, \mathcal{T}, c). \quad (4)$$

IV. MOTIONGPT-2: A VERSATILE MODEL SUPPORTING HOLISTIC MOTION UNDERSTANDING AND GENERATION

Building upon the MotionGPT, MotionGPT-2 (as shown in Figure. 2) is to formulate a **multi-task unified** (motion generation, motion captioning, and generalized motion completion) framework to support **holistic motion representations** (fine-grained body and hand movements). The key designs are as follows: (1) *Part-Aware VQ-VAE for Body-hand Tokenization*. By designing an innovative Part-Aware VQVAE, our MotionGPT-2 achieves informative and compact representations of fine-grained holistic motions, making it adaptable to holistic motion generation tasks (Section IV-A). (2) *A Unified Motion-Language Vocabulary*. By expanding the text vocabulary of LLM, we develop a unified Large Motion-Language Model (LMLM) (Section IV-B) for the seamless integration of motion and language modalities. This vocabulary extension enables MotionGPT-2 to support a broader range of novel motion-related tasks by different instructions (Section IV-C). (3) *Three-Stage Training Strategy*. To enhance alignment between motions and texts, we develop a three-stage training pipeline (Section IV-D), including Motion Tokenization, Motion-Language Alignment, and Instruction Fine-tuning.

A. Part-Aware VQVAE for Holistic Motion Representations

Our previous MotionGPT is limited to the generation of body-only motions. Yet, advancing toward holistic motion

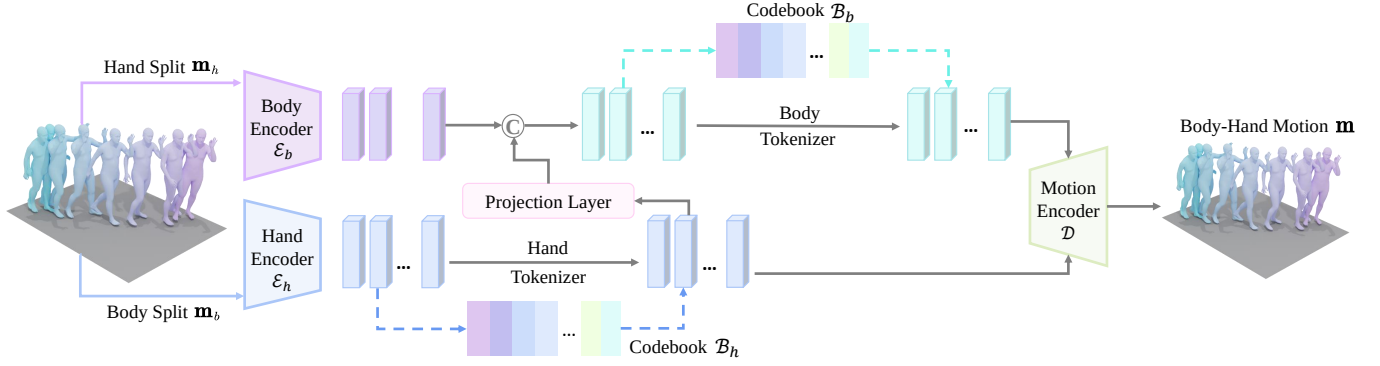


Fig. 3: The framework overview of our proposed Part-Aware VQVAE for body-hand motion tokenization. The Part-Aware VQVAE splits SMPL-X-based human representations into body-hand motions $\mathbf{m}_b$ and $\mathbf{m}_h$. Then, it quantizes fine-grained body-hand motion into two discrete codebooks $\mathcal{B}_b$ and $\mathcal{B}_h$ with hierarchical body priors.

generation is essential for achieving more realistic and lifelike animations. Holistic motions are considerably more complex than their body-only counterparts, necessitating the detailed and coordinated motion generation involving the body and hand features. In this section, our MotionGPT-2 emphasizes the representation of body-hand motions. With the above single motion vocabulary approach, the embedding of holistic human motion inevitably leads to ambiguities where similar actions are represented by the same motion token. To model distinct semantic granularity for body and hand, we design a Part-Aware VQVAE module, which utilizes the codebook $\mathcal{B}_b$ and $\mathcal{B}_h$ to learn discrete representations for body and hand, in a more structured and coordinated manner.

As illustrated in Fig. 3, we input the SMPL-X based body-hand representations $\mathbf{m}^B = \{m_1^B, m_2^B, \dots, m_T^B\} \in \mathbb{R}^{T \times d}$ and $\mathbf{m}^H = \{m_1^H, m_2^H, \dots, m_T^H\} \in \mathbb{R}^{T \times d}$ into two separate encoders. As shown in Fig. 3, the hand embedding $\mathbf{e}_h$ is first quantized by the hand codebook $\mathcal{B}_h$ and we then fuse the body and hand tokens via the concatenation operator before quantizing the body embedding $\mathbf{e}_b$ using the body codebook $\mathcal{B}_b$. The interaction of body and hand motions is tightly coupled and articulated, shaped by biomechanical and physical restrictions in the context of complex activities. Such a *fuse-before-quantize* method for body tokens effectively enhances the overall naturalness and coordination of holistic motion representations. Akin to the [67], the Body-hand Decoder is designed to take body tokens p^B and hand tokens p^H as input to accurately reconstruct corresponding motions.

B. A Unified Motion-Language Vocabulary

Most previous motion-related studies [1], [24], [29], [63], [68] have viewed textual descriptions and motions as separate modalities. However, similar to sentences in natural language, a continuous motion $\mathbf{m}$ is compressed into motion tokens, serving as a type of “body language”. Therefore, our MotionGPT-2 naturally uses LLMs to jointly model motion and language representations. It reflects how humans execute body motions, allowing for seamless transitions across motion-related modalities in real-life scenarios.

Specifically, to merge multi-modal discrete tokens with the pre-trained LLM, the original vocabulary $\mathcal{B}_t$ is expanded to incorporate motion tokens $\mathcal{B}_m$. Further, we also insert several

special tokens $\mathcal{B}_s$ into the LLM’s vocabulary, *e.g.*, $\langle \text{motion} \rangle$ and $\langle / \text{motion} \rangle$, which signify the beginning and end of the motion. This extends the vocabulary into a unified text-motion set, $\mathcal{B} = \{\mathcal{B}_t, \mathcal{B}_m, \mathcal{B}_s\}$. The newly incorporated parameters are initialized randomly. Expanding the LLM’s vocabulary, a human motion is denoted as a token sequence that is LLM-understandable. Equipped with the vocabulary $\mathcal{B}$, we model the joint distribution of textual descriptions and human motions in a unified space. It allows us to formulate motion-related tasks in a general framework, leveraging the task-aware instructions.

C. Instruction Prompts in New Motion-Related Tasks

Our previous MotionGPT primarily focused on motion generation tasks. In contrast, our MotionGPT-2 enhances its capabilities by defining several core motion-related tasks, including motion captioning, motion prediction, and motion interpolation. For each task, we construct unique instruction prompts tailored to the specific task requirements. For instance, given a LLM $\mathcal{F}$, the instruction template $\mathcal{I}$ for motion captioning and motion prediction task as well as the answer of the LLM $\hat{\mathcal{P}} = \mathcal{F}(\mathcal{I})$ are defined as:

Below is an instruction that describes a task, paired with an input that provides further context. Write a response that appropriately completes the request.

% Task Prompts: Motion Captioning Task Prompts
% Control Conditions: Code Sequences of Control Conditions (Motion Tokens)
Instruction $\mathcal{I}$: {Task Prompts $\mathcal{T}$ } {Control Conditions}
Answer $\hat{\mathcal{P}}$: {Sequences of Text Tokens}

% Task Prompts: Motion Prediction Task Prompts
% Control Conditions: Code Sequences of Control Conditions (Initial Several Motion Tokens)
Instruction $\mathcal{I}$: {Task Prompts $\mathcal{T}$ } {Control Conditions}
Answer $\hat{\mathcal{P}}$: {Sequences of Human Motion Tokens}

In the text-based holistic motion generation task, the instruction tuning phase proceeds after obtaining the discrete motion tokens p^B and p^H . During this stage, we formulate the instruction template $\mathcal{I}$ and the answer $\mathcal{P}$ of the LLM for motion generation. Inspired by MLLMs [31], [33], we unify the discrete body-hand tokens within a general prompt template.

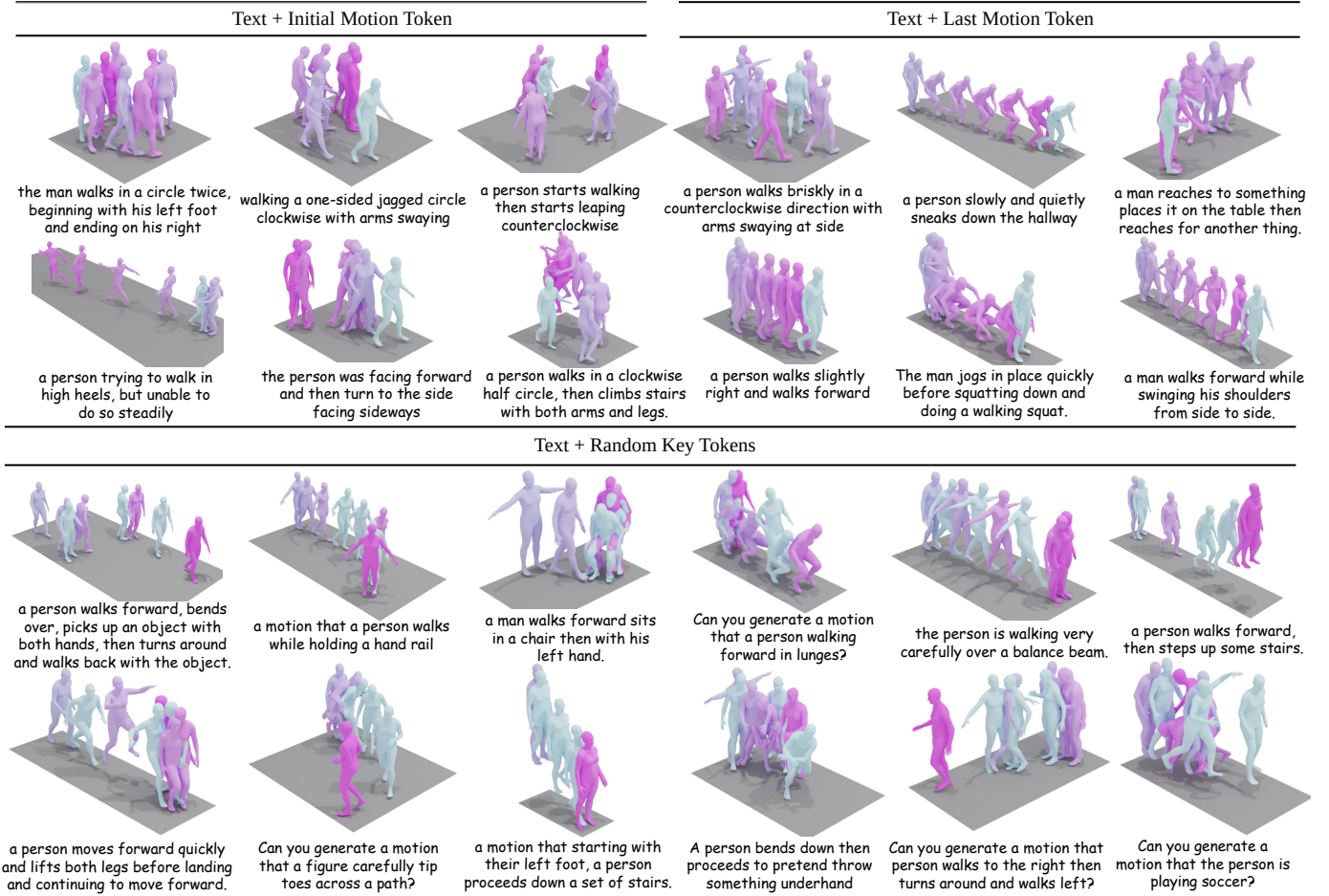


Fig. 4: Gallery showcasing the results of generated human motions by MotionGPT-2 with multiple control conditions on the HumanML3D dataset [30], i.e., *Text+Initial Motion Token*, *Text+Last Motion Token*, and *Text+Random Key Motion Token*. With these diverse control signals, our MotionGPT-2 demonstrates the ability to generate physically realistic human motions.

Below is an instruction that describes a task, paired with an input that provides further context. Write a response that appropriately completes the request.

% **Task Prompts:** Motion Generation Task Prompts $\mathcal{T}$
 % **Control Conditions:** Sequences of Control Conditions (Text Tokens)

Instruction $\mathcal{I}$: {Task Prompts $\mathcal{T}$ } {Control Conditions}
Answer $\hat{\mathcal{P}}$: {< Motion_Body_Token >< Body_Token >< /Motion_Body_Token >< Motion_Hand_Token >< /Motion_Hand_Token >< /Motion_Body_Token >}

D. Three-Stage Training Strategy

Our previous MotionGPT [1] is founded on a two-stage approach. Nevertheless, the absence of motion-language alignment hinders model from interpreting the complex relationships between motions and texts. Towards this issue, MotionGPT-2 proposes a comprehensive three-stage training strategy:

Stage 1: Training of Motion Tokenizer. As depicted in Section III-A, we first learn the discrete motion representations to align with the LLM’s token-in-token-out nature. Guided by the objective outlined in Eq. 3, the quantization process allows human motions $\mathbf{m}$ to be expressed as a sequence of tokens, which seamlessly integrates with descriptive text. To maintain

stability in LLM optimizing, we subsequently freeze weights of motion-aware VQVAE during further training stages.

Stage 2: Motion-Language Alignment. Inspired by current Multi-modal Large Language Models (MLLMs) [31], [33], to align the motion and language feature space of MotionGPT-2, the LLaMA [44] model is fine-tuned on a wide range of motion-related tasks. *To accurately interpret contextual ‘body language’ semantics while preserving the text generation capability of the LLM.* We finetune the LLM with LoRA using a mixture of language and motion data in both unsupervised and supervised manners. Similar to LLaVA [33], we learn the complex relationship of language and motions with the paired text-motion datasets. Further, we ensure the text generation capability through the LLM’s next-token prediction mechanism on text-only data. Compared with our conference MotionGPT [1], with the additional motion-language alignment stage, MotionGPT-2 offers improved semantic consistency and fine-grained control of human motions and textual descriptions.

Stage 3: Fine-tuning LLM by Motion Instructions. Instruction tuning enables LLMs to handle various motion generation and understanding tasks by asking LLM questions in different instructions. Hence, we devise a set of instructions that merge task descriptions with control signals and employ the efficient LoRA [36] technique to fine-tune LLMs.

TABLE I: **Quantitative results of text-based motion generation on the HumanML3D dataset.** “Real” denotes the results computed with GT motions. “ $\rightarrow$ ” indicates metrics that are better when closer to “Real” distribution. “MultiModal Dist.” denotes the Multi-Modality Distance. We conduct each evaluation 20 times, presenting the average metric and a 95% confidence interval, with the top scores marked in bold.

Model	Top 1	R-Precision $\uparrow$ Top 2	Top 3	FID $\downarrow$	MultiModal Dist. $\downarrow$	Diversity $\rightarrow$	MultiModality $\uparrow$
Real	0.511 $\pm$.003	0.703 $\pm$.003	0.797 $\pm$.002	0.002 $\pm$.000	2.974 $\pm$.008	9.503 $\pm$.065	—
TM2T [63]	0.424 $\pm$.003	0.618 $\pm$.003	0.729 $\pm$.002	1.501 $\pm$.046	3.467 $\pm$.008	8.589 $\pm$.058	2.424 $\pm$.093
T2M [30]	0.457 $\pm$.002	0.639 $\pm$.003	0.740 $\pm$.003	1.067 $\pm$.002	3.340 $\pm$.008	9.188 $\pm$.002	2.090 $\pm$.083
MDM [69]	0.319 $\pm$.005	0.498 $\pm$.004	0.611 $\pm$.007	0.544 $\pm$.001	5.566 $\pm$.027	9.559 $\pm$.086	2.799 $\pm$.072
MotionDiffuse [17]	0.491 $\pm$.001	0.681 $\pm$.001	0.782 $\pm$.001	0.630 $\pm$.001	3.113 $\pm$.001	9.410 $\pm$.049	1.553 $\pm$.042
MLD [25]	0.481 $\pm$.003	0.673 $\pm$.003	0.772 $\pm$.002	0.473 $\pm$.013	3.169 $\pm$.010	9.724 $\pm$.082	2.413 $\pm$.079
T2M-GPT [24]	0.491 $\pm$.003	0.680 $\pm$.003	0.775 $\pm$.002	0.116 $\pm$.004	3.118 $\pm$.011	9.761 $\pm$.081	1.856 $\pm$.011
MoMask [26]	0.521 $\pm$.002	0.713 $\pm$.002	0.807 $\pm$.002	0.045 $\pm$.002	2.958 $\pm$.008	9.620 $\pm$.064	1.241 $\pm$.040
ReMoDiffuse [29]	0.510 $\pm$.005	0.698 $\pm$.006	0.795 $\pm$.004	0.103 $\pm$.004	2.974 $\pm$.016	9.018 $\pm$.075	1.795 $\pm$.043
AttT2M [70]	0.499 $\pm$.003	0.690 $\pm$.002	0.786 $\pm$.002	0.112 $\pm$.006	3.038 $\pm$.007	9.700 $\pm$.090	2.452 $\pm$.051
GraphMotion [71]	0.504 $\pm$.003	0.699 $\pm$.002	0.785 $\pm$.002	0.116 $\pm$.007	3.070 $\pm$.008	9.692 $\pm$.067	2.766 $\pm$.096
MotionGPT [1]	0.364 $\pm$.005	0.533 $\pm$.003	0.629 $\pm$.004	0.805 $\pm$.002	3.914 $\pm$.013	9.972 $\pm$.026	2.473 $\pm$.041
MotionGPT [27]	0.492 $\pm$.003	0.681 $\pm$.003	0.733 $\pm$.006	0.232 $\pm$.008	3.096 $\pm$.008	9.528 $\pm$.071	2.008 $\pm$.084
MotionLLM [65]	0.482 $\pm$.004	0.672 $\pm$.003	0.770 $\pm$.002	0.491 $\pm$.019	3.138 $\pm$.010	9.838 $\pm$.244	—
MotionGPT-2 (Ours)	0.496 $\pm$.002	0.691 $\pm$.003	0.782 $\pm$.004	0.191 $\pm$.004	3.080 $\pm$.013	9.860 $\pm$.026	2.137 $\pm$.022

V. EXPERIMENTS

A. Experimental Setup

The HumanML3D dataset [30] stands as the largest available dataset focused solely on 3D body motion and associated textual descriptions. It comprises 14,616 motion clips paired with 44,970 meticulously annotated descriptions derived from a vocabulary of 5,371 unique words. These motion sequences are sourced from the AMASS [72] and HumanAct12 [55] motion capture collections, showcasing a diverse array of human activities, including everyday tasks, sports, acrobatics, and artistic expressions. Each motion clip is linked with 3-4 descriptive texts. For training, one sentence is randomly chosen as the match, while for testing, the first text description is consistently used to evaluate model performance. The motion clips are down-sampled to 20 FPS and vary in duration from 2 to 10 seconds. The dataset is divided into training, validation, and test subsets, allocated in an 80%, 5%, and 15% ratio, respectively, with no overlaps among them.

The KIT-ML [37] dataset is comprised of 3,911 motion sequences along with 6,278 textual descriptions. Each sequence is associated with one to four sentences, with the average description containing 9.5 words. This dataset merges selected elements from the KIT WholeBody Human Motion Database [73] and the CMU Graphics Lab Motion Capture Database [74], with an emphasis on locomotion motions. The motion sequences in KIT-ML have been down-sampled to a frame rate of 12.5 fps.

The Motion-X [38] dataset is currently the largest whole-body expressive motion repository, comprising 95,642 high-fidelity 3D motion sequences based on the SMPL-X model [75], paired with corresponding pose descriptions and semantic labels for each sequence. The Motion-X dataset gathers 15K monocular videos from various online sources and public video dataset, capturing a wide range of scenarios such as daily actions, sports activities, and many domain-specific scenes,

TABLE II: Assessment of motion generation across conditions. With initial/key tokens, MotionGPT(†) and MotionGPT-2(‡) show superior performance compared to the text-only version.

Methods	FID $\downarrow$ Dist. $\downarrow$ Diversity $\uparrow$	FID $\downarrow$ Dist. $\downarrow$ Diversity $\uparrow$
	HumanML3D(†)	HumanML3D(‡)
Text-Only	0.567 3.775 9.006	0.191 3.080 9.860
Text + Initial Pose	0.520 3.844 9.588	0.183 3.285 10.066
Text + Last Pose	0.591 3.718 9.251	0.358 3.673 9.582
Text + Random Pose	0.367 3.598 9.176	0.182 3.031 10.102
	KIT-ML(†)	KIT-ML(‡)
Text-Only	0.597 3.394 10.540	0.614 3.164 11.256
Text + Initial Pose	0.664 3.445 10.390	0.756 3.362 11.053
Text + Last Pose	0.856 3.336 10.580	0.784 3.483 11.460
Text + Random Pose	0.671 3.411 10.760	0.807 3.173 11.447

with 13.7M frame-level 3D whole-body pose annotations. In this paper, we standardize the Motion-X dataset by selecting 52 joints from the human body and hands.¹

B. Implementation Details

Motion Data Pre-processing. We follow the dataset pre-processing procedures outlined in [1], [24], [30]. The raw 3D motion coordinates are first aligned with a default human skeletal model. The Y-axis is then set perpendicular to the ground, allowing individuals to face the Z+ direction. These coordinates are then processed into motion features, including foot contact, global rotations and translations, local joint positions, velocities, and 6D rotations. The final dimensions are 263 for the HumanML3D dataset, 251 for the KIT-ML dataset, and 623 for the Motion-X dataset. For SMPL-based motion representations, the maximum motion length is set to 196, with minimums of 40 for HumanML3D and 24 for KIT-ML. In

¹More experiments results and visualizations on the KIT-ML benchmark are available in the *Supplementary Materials*.

TABLE III: Experiments of motion captioning task on the HumanML3D [30] benchmark. Results marked with * are from MotionGPT [27], and were computed using unprocessed ground truth texts for linguistic metrics.

Methods	R Precision $\uparrow$			MultiModal Dist. $\downarrow$	Length $_{avg}\uparrow$	Bleu1 $\uparrow$	Bleu4 $\uparrow$	Rouge $\uparrow$	Cider $\uparrow$	BertScore $\uparrow$
	Top1	Top2	Top3							
Real Desc	0.523	0.725	0.828	2.901	—	—	—	—	—	—
RAEs	0.100	0.188	0.261	6.337	—	33.3	10.2	37.5	22.1	10.7
Seq2Seq(Att)	0.436	0.611	0.706	3.447	—	51.8	17.9	46.4	58.4	29.1
SeqGAN	0.332	0.457	0.532	4.895	—	47.8	13.5	39.2	50.2	23.4
TM2T*	0.516	0.720	0.823	2.935	10.67	48.9	7.00	38.1	16.8	32.2
MotionGPT*	0.543	—	0.827	2.821	13.04	48.2	12.5	37.4	29.2	32.4
MotionGPT-2 (Ours)	0.558	0.738	0.838	2.767	15.27	48.7	13.8	37.6	29.8	32.6

TABLE IV: Ablations on the effects of LLM types and scales on text-based motion generation, evaluated on the HumanML3D [30] benchmark. In addition to full fine-tuning of the T5-base model, LoRA is used for optimizing other decoder-only LLMs.

Model	Trainable Parameters	R-Precision $\uparrow$			FID $\downarrow$	MultiModal Dist. $\downarrow$	Diversity $\rightarrow$	MultiModality $\uparrow$
		Top 1	Top 2	Top 3				
Real	—	0.511	0.703	0.797	0.002	2.974	9.503	—
T5-base (Pretrain)	Full-Finetune	0.314	0.457	0.554	0.493	4.662	9.634	1.793
T5-base (Finetune)	Full-Finetune	0.417	0.581	0.668	0.148	3.989	9.961	1.985
Gemma-2b-It (Pretrain)	34M	0.385	0.546	0.643	0.207	4.559	9.733	2.351
Gemma-2b-It (Finetune)	34M	0.436	0.600	0.697	0.228	3.589	10.081	2.269
Gemma-7b-It (Pretrain)	101M	0.404	0.575	0.673	0.219	3.768	9.964	2.364
Gemma-7b-It (Finetune)	101M	0.446	0.622	0.715	0.177	3.545	9.652	2.198
LLaMA3-8B (Pretrain)	89M	0.438	0.619	0.717	0.314	3.514	10.010	2.252
LLaMA3-8B (Finetune)	89M	0.482	0.668	0.760	0.282	3.288	10.212	2.261
LLaMA3.1-8B (Pretrain)	89M	0.456	0.630	0.732	0.291	3.417	9.884	2.145
LLaMA3.1-8B (Finetune)	89M	0.496	0.691	0.782	0.191	3.080	9.860	2.137

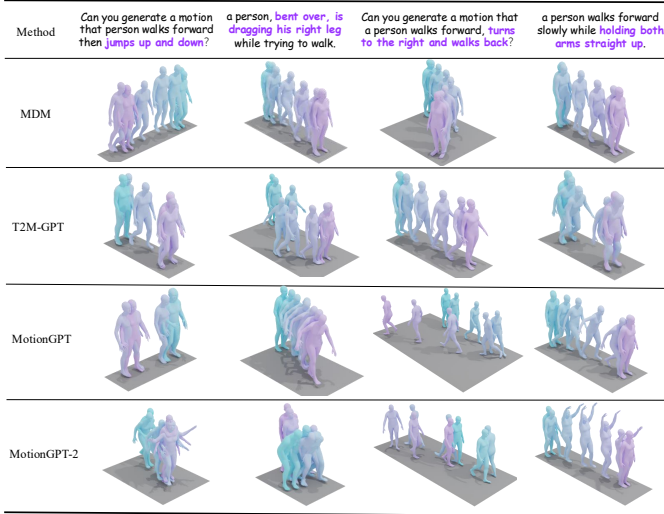


Fig. 5: Showcase of visualization results for the text-based motion generation task using the HumanML3D [30] dataset. We compare our MotionGPT-2 with the state-of-the-art method, i.e., MDM [69], T2M-GPT [24], MotionGPT [1].

SMPL-X-based Motion-X dataset, the maximum motion length frames is 300, with a minimum of 40.

Training Details. In the following experiments, we adopt the decoder-only LLaMA 3.1-8B [44] model as the foundational LLM, keeping its parameters frozen while applying fine-tuning through the LoRA [36] method. The motion tokenizer is trained over 1,200 epochs, with an initial learning rate of 1×10^{-4} . We set the mini-batch size to 256 and use the

AdamW optimizer [76], with a weight decay of 1×10^{-5} for model optimization. Following previous studies [24], [27], we set the codebook $\mathcal{B}_m \in \mathbb{R}^{512 \times 512}$ for motion VQ-VAE and the codebook $\mathcal{B}_b \in \mathbb{R}^{512 \times 512}$ and $\mathcal{B}_h \in \mathbb{R}^{512 \times 512}$ for Part-Aware VQ-VAE. The motion encoder applies a temporal down-sampling rate of 4. Our MotionGPT-2 leverages a learning rate of 2×10^{-4} during the pre-training phase and 1×10^{-4} in the instruction tuning phase of the unified motion-language learning process. The mini-batch size is 32. The pre-trained LLM undergoes 100 epochs in the *Motion-Language Alignment* phase and 50 epochs during the *Instruction Tuning* phase. MotionGPT-2 and the re-implemented models are built using PyTorch, with all experiments on 4 NVIDIA 80G A100 GPUs.

C. Results on Human Motion Generation

Text-based Body-only Motion Generation. The quantitative results of motion quality are depicted in Table I. Among these, Table I provide quantitative comparisons on the SMPL-based HumanML3D [30] dataset and the KIT-ML [73] dataset. The results presented reflect the performance of MotionGPT-2, which has been pre-trained across multiple motion-related tasks and subsequently fine-tuned for the specific text-based motion generation task. By tuning mere 1% of LLM parameters, our general-purpose MotionGPT-2 exhibits a performance that is competitive with state-of-the-art approaches. Compared to our previous version, the MotionGPT-2 yields a significant **10.4%** improvement in R-Precision Top-3 and a **0.254** reduction in FID score on HumanML3D. Through the tailored *Motion-Language Alignment* stage, our MotionGPT-2 exhibits greater semantic consistency with the textual description, improving interpretation of body language semantics. Further, compared

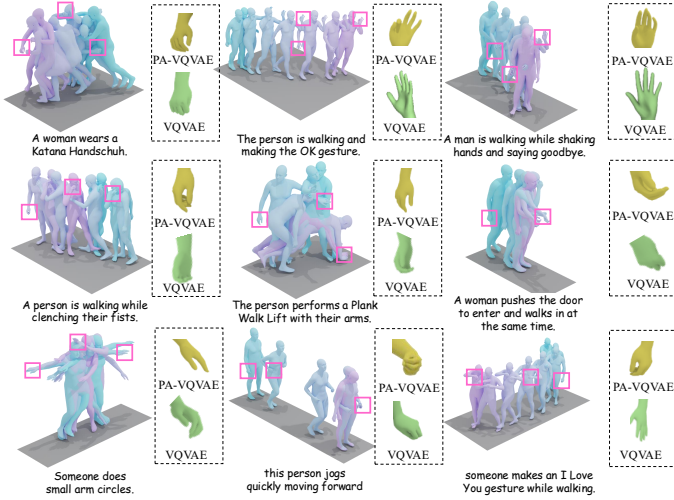


Fig. 6: Qualitative results of our method on Motion-X [38]. Utilizing world knowledge of LLMs, our PA-VQVAE demonstrates the capability to effectively capture lifelike hand interactions.

Initial token	Recon MotionGPT	Vel MotionGPT	Recon MotionGPT-2	Vel MotionGPT-2
Text-only	24.70	1.095	19.19	0.846
Text + Initial poses	13.78	0.549	7.59	0.328
Last token	MotionGPT	MotionGPT-2	MotionGPT	MotionGPT-2
Text-only	19.70	1.172	11.77	0.735
Text + Last poses	6.831	0.397	4.376	0.291
Key tokens	MotionGPT	MotionGPT-2	MotionGPT	MotionGPT-2
Text-only	8.035	3.813	6.591	1.932
Text + Random poses	5.383	2.423	4.139	1.055

TABLE V: Evaluation of the consistency with pose control conditions on the HumanML3D [30] test sub-set using the pre-trained LLaMA 3.1-8B model.

to other LLM-based methods, our MotionGPT-2 fine-tunes the LLM and relate world knowledge to 3D human motions, achieving superior generation beyond existing solutions.

As shown in Fig. 5, we conduct visualization experiments on the text-based motion generation task to vividly illustrate the capabilities of our MotionGPT-2 model. In Fig. 5, we observe that the diffusion-based MDM [69] method generates fewer semantic motions aligned with the provided descriptions. By utilizing discrete motion tokens, T2M-GPT [24] can better learn motion patterns and semantics which leads to more coherent and contextually rich motions. Our conference version MotionGPT [1], with fine-tuned LLMs, generalizes well to more diverse and complex text prompts. In contrast to MotionGPT, our MotionGPT-2 extends motion vocabulary to original LLMs for jointly modeling motion and language representations in a unified space.

Multiple Signal Controlled Body-only Motion Generation.

Besides text inputs, MotionGPT-2 is capable of incorporating human poses as an additional control modality, with the resulting motion quality detailed in Table II. Introducing extra controls, such as initial, last, or key poses, does not degrade motion quality. In fact, MotionGPT-2 shows superior

Input Motions	Real	MotionGPT	Ours
	a person walks forward, picks something up off of the floor, then puts it on a low shelf or object.	a person is walking forward, grabs something, turns around, and sit the object down.	a person walks forward, they stop and bend down to pick something up , then continue walking.
	the toon is walking in a zig zag motion, shuffling a bit at the end of the "plane"	this person walks in a zig zag motion from back to front	a man strolls slowly , and staggers to the left in a zig zag pattern
	a person throws something , then catches something, then throws it	a person throws something and then catches something.	person looks like he is throwing a baseball then catches one and throws it again
	person walks forward and curves to the left.	person is standing still, slightly shifting their weight to the left.	a person walks in an arc to the left then stops.

Fig. 7: Comparison of other methods on motion captioning task. The results demonstrate that our MotionGPT-2 generates more conceptually and semantically rich motion descriptions. Specific words are marked to highlight the semantic similarity of the generated captions and the real one.

performance when initial or key tokens are provided, achieving FID scores of **0.183** or **0.182**, an improvement over the text-only model's 0.191 on the HumanML3D benchmark, demonstrating its flexibility in handling diverse control modalities. In fact, MotionGPT-2 shows superior performance when initial or key tokens are provided, achieving FID scores of **0.183** or **0.182**, an improvement over the text-only model's 0.191 on the HumanML3D benchmark. In spite of this, MotionGPT-2's performance is still notable, reinforcing its proficiency in generating high-quality and diverse motions across a range of control conditions. As demonstrated in Fig. 4, the motions produced by our model align closely with the specified poses and consistently follow the textual descriptions.

Holistic Motion Generation. In this part, we focus on the holistic motion generation task. As shown in Figure 9, we evaluate various LLMs on the SMPL-X-based Motion-X [38] dataset. Our tailored PA-VQVAE consistently outperforms the original VQVAE across multiple scales and types of LLMs, demonstrating the effectiveness of our innovative motion discretization framework. For instance, compared to its VQVAE counterpart, Part-Aware VQVAE (LLaMA 3.1-8B) achieves an improvement in Top-1 R-Precision from 0.332 to 0.349, and a decrease in FID from 0.666 to 0.619. The superior performance of Part-Aware VQVAE over the original VQVAE can be attributed to its more fine-grained discretization representation and hierarchical modeling capability for human motion. Moreover, utilizing distinct motion vocabularies for body and hand reduces the ambiguity, where similar actions could be represented by the same token. As shown in Fig. 6, our MotionGPT-2 is capable of generating vivid motions that accurately correspond to the given text descriptions.

Perceptual Studies. We assess visual realism and semantic fidelity of generated 3D motions by rendering 2,000 representative clips from multiple camera views and evaluating them with Qwen2.5-VL-72B [77], which scores each sample across five dimensions: **Naturalness**, **Semantic Consistency**, **Fluency**, **Detail**, and **Overall Preference** (definitions in the supplementary material). In Figure 8, MotionGPT-2 consistently surpasses three strong baselines (MotionGPT, Gemma-2B-It, Gemma-7B-It) across all dimensions, demonstrating its

TABLE VI: Comparisons between separate training for each task and joint training for multiple tasks on HumanML3D [30] test set using the LLaMA 3-8B model. We use orange and green markings to represent decrements and improvements in the metric, respectively. Joint training can achieve better performance for all tasks.

Task	Training Strategy	FID ↓	MultiModal Dist.↓	Top-1	R-Precision ↑ Top-2	Top-3	Diversity ↑
Text	Separate	0.523	3.627	0.358	0.514	0.604	9.108
+ Initial token		0.483	3.489	0.378	0.549	0.647	9.614
+ Last token		0.974	4.208	0.339	0.501	0.598	9.598
+ Key tokens		0.428	3.276	0.424	0.617	0.697	9.929
Text	Joint	0.482(-0.041)	3.295(-0.332)	0.419(+0.061)	0.597(+0.083)	0.683(+0.079)	9.422(+0.314)
+ Initial token		0.454(-0.029)	3.173(-0.316)	0.434(+0.056)	0.613(+0.064)	0.710(+0.063)	9.573(-0.041)
+ Last token		0.507(-0.467)	3.860(-0.348)	0.427(+0.088)	0.608(+0.107)	0.723(+0.125)	9.688(+0.090)
+ Key tokens		0.406(-0.022)	3.459(-0.183)	0.445(+0.021)	0.616(-0.001)	0.722(+0.025)	9.987(+0.058)

TABLE VII: Evaluation of motion prediction and in-betweening on part of the AMASS [72], considering only motion data. FID reflects the quality of the generated motions, while Diversity quantifies the motion variability within each condition.

Methods	Motion Prediction				Motion In-between		
	FID ↓	Diversity ↑	ADE ↓	FDE ↓	FID ↓	Diversity ↑	ADE ↓
Real	0.002	9.503	—	—	0.002	9.503	—
MDM	6.031	7.813	5.446	8.561	2.698	8.420	3.787
MotionGPT	0.905	8.972	4.745	6.040	0.214	9.560	3.762
MotionGPT-2	0.537	9.414	4.512	5.823	0.408	9.327	3.704

clear perceptual superiority. MotionGPT-2 attains the highest mean scores in Naturalness (3.84) and Fluency (3.45), reflecting physically coherent and temporally smooth motion generation enabled by Part-Aware VQ-VAE. Its Semantic Consistency (3.74) highlights accurate grounding between texts and motion dynamics through instruction tuning. In Fine-Grained Detail (3.16), the model captures subtle hand and body gestures, a long-standing challenge for prior methods. Finally, the Overall Preference (4.11)—obtained via pairwise comparison—confirms MotionGPT-2’s clear human-perceived advantage, with over 60% of evaluators preferring its outputs. Together, these results provide strong human-centered evidence that the proposed PA-VQVAE and LLM designs of MotionGPT-2 enhance motion quality, realism, and semantic fidelity.

User Study for Fine-Grained Hand Motion Evaluation.

We conduct a small-scale user study involving 10 participants with backgrounds in computer vision or human motion analysis. Each participant is presented with 20 paired video clips generated by the vanilla VQ-VAE and our Part-Aware VQ-VAE under identical text prompts (e.g., “making the OK gesture”, “playing the piano”). In each pair, the participants are asked to select the sequence that appears more natural in terms of fine-grained hand articulations. Across the total of 200 pairwise comparisons, 81.5% of the selections favor the outputs of PA-VQVAE. This strong and consistent preference indicates that our hierarchical design not only improves objective reconstruction metrics but also yields perceptible gains in the naturalness and plausibility of hand motions. These results provide compelling subjective evidence, complementing our quantitative evaluations, and directly address the reviewer’s suggestion to assess hand motion quality through user studies.

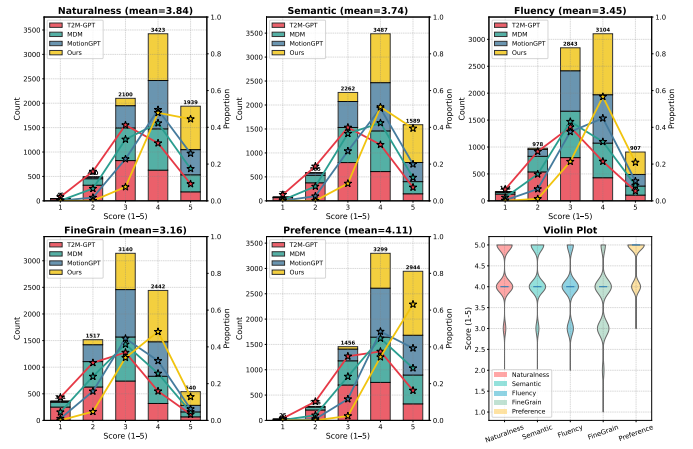


Fig. 8: **Subjective Assessment of Motion Quality.** A subjective assessment of our model’s performance compared to T2M-GPT, MDM, and MotionGPT across five qualitative metrics.

D. Results on Motion Captioning

Compared to the text-to-motion task, the motion-to-text task involves generating a textual description from a given human motion sequence. As in [27], we rely on ground truth descriptions for more accurate assessment. Table III presents the results of quantitative evaluation for motion-to-text translation on the HumanML3D dataset. The comparisons in Table III shows that our proposed MotionGPT-2 outperforms recent works in generating text descriptions for the given motions. The generated language descriptions deliver substantial improvements in both linguistic quality (BLEU [78] and BertScore [79]) and the precision of motion retrieval (R-Precision). By fine-tuning LLMs, our MotionGPT-2 emerges as a specialized tool endowed with extensive world knowledge, thereby enhancing its capacity to interpret human motion.

In Fig. 7, we provide further examples of translating motion to text using the HumanML3D dataset. All methods are evaluated under the same training and inference conditions on HumanML3D. Compared to MotionGPT [27], our MotionGPT-2, which leverages LLM-interpretable motion tokens, is capable of generating more conceptual and semantically rich motion descriptions. For instance, it can produce descriptions such as “walk in an arc shape” and “throwing a baseball”, offering more intuitive understanding of the given motion motions.

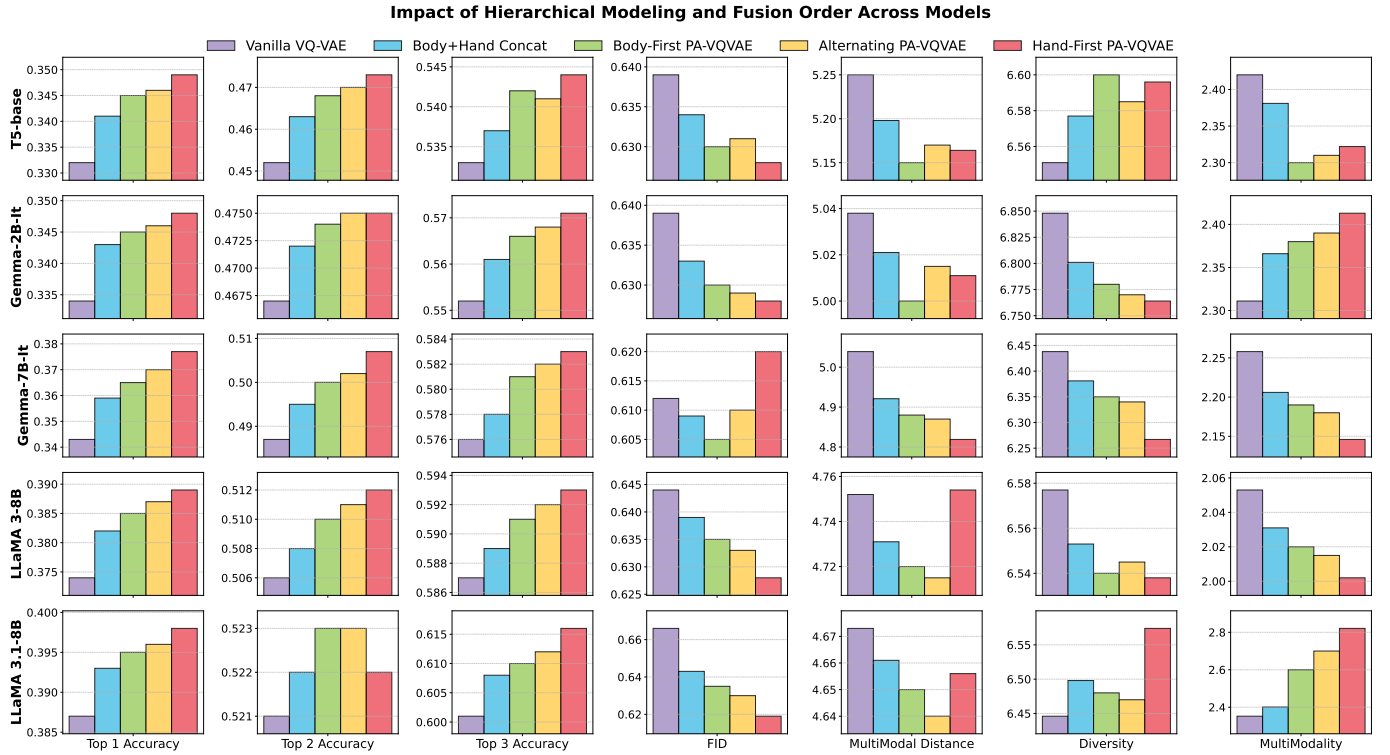


Fig. 9: **Impact of Hierarchical Modeling and Fusion Order on Motion Generation Performance.** This ablation study across five LLM backbones demonstrates that our Hand-First PA-VQVAE consistently outperforms Vanilla VQ-VAE, Body+Hand Concat, Body-First, and Alternating variants across key metrics (Top-1/2/3 Accuracy, FID, Diversity, MultiModality).

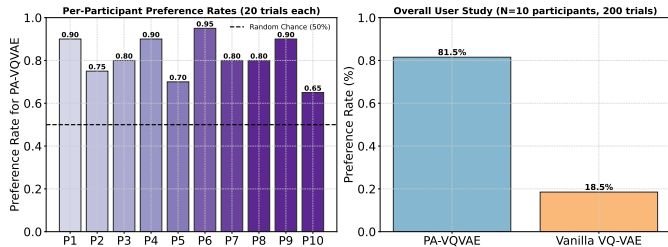


Fig. 10: **Subjective Evaluation of Hand Motion Fidelity.** (Left) Individual preference rates from a 10-participant study comparing PA-VQVAE and Vanilla VQ-VAE under identical prompts. (Right) Aggregated outcomes reveal an 81.5% overall preference for PA-VQVAE across 200 trials.

E. Results on Generalized Motion Completion

Following [27], we classify both motion prediction and in-betweening together as generalized motion completion in Table VII. For the motion prediction task [80]–[82], only the first 20% of the sequence is used as the conditioning input, while approximately 50% of the motion is intentionally masked at random to assist in the completion process. Similar to text-based motion generation, as shown in Table VII, we also fine-tune the proposed MotionGPT-2 specifically for this specific task and utilize FID, ADE, and FDE as metrics. The unified motion-language framework of the MotionGPT-2 leverages contextual information of fine-tuned LLM to understand in-depth motion dynamics. Compared to [27], [69], our MotionGPT-2 demonstrates a remarkable capability to

generate contextually appropriate motion completions.

F. Ablation Study

Capability of Pre-trained LLM. As shown in Table IV, we delve into how different scales and types of LLMs affect the performance of text-based human motion generation tasks on the HumanML3D [30] and KIT-ML [73] datasets. We observe that: (1) Larger LLMs (*e.g.*, LLaMA 3.1-8B and LLaMA 3-8B) offer distinct advantages over smaller counterparts, achieving significant improvements in fidelity (FID) and multimodal alignment (*i.e.*, R Precision, MultiModal Dist.). To explain, the improved context understanding of LLMs ensures that the output motion aligns closely with intended actions. Further, LLMs with comprehensive world knowledge can synthesize physically plausible motions even when faced with linguistic ambiguity. (2) LLMs fine-tuned by unified instructions demonstrate clear advantage in maintaining semantic consistency and producing motions that are better aligned with textual descriptions. For instance, the fine-tuned Gemma-7B-It outperforms its pre-trained counterpart, achieving a 10% improvement in R-Precision Top-3 (from 0.673 to 0.715) and a 19% reduction in FID (from 0.219 to 0.177). (3) Compared to T5-base used in [27], [28], which requires full fine-tuning, fine-tuning the decoder-only LLaMA 3.1-8B yields a higher R-Precision Top-3 (0.782) and a lower FID (0.191). This indicates that models with more parameters possess capacity to capture complex relationships between text and motion.

Consistency with pose control conditions. We assess the benefits of pose control by comparing the consistency between the controlled poses and the generated motions on

the HumanML3D test sub-set. Concerning each specific task (initial/last/key), motions are generated both with and without pose controls, utilizing the (text+pose)-to-motion and text-to-motion methods, respectively. The results, displayed in Tab. V indicate that key-frame consistency is higher with pose controls than in text-only generation counterpart, proving the effectiveness of (text+pose)-to-motion with pose control. Such results highlight the critical role that pose controls play in coherent and contextually appropriate human motion synthesis.

Comparison with Separate Training. We carry out task-specific training on the HumanML3D dataset [30] to test the effectiveness of the proposed MotionGPT-2 model in motion generation. This experimental setup is implemented to determine whether a multi-task learning framework can enhance the performance of each distinct control condition independently. The comparison results are presented in Table VI. Our findings indicate that joint training across all tasks significantly improves performance metrics across the board. Notably, this enhancement is particularly pronounced when utilizing text and the last pose token as input conditions. These results illustrate the value of our multi-modal signals controlled motion generation. MotionGPT-2’s ability to generate motions under a specific input condition is strengthened by drawing knowledge from other conditions.

Impact of Hierarchical Modeling and Fusion Order. We conduct comprehensive ablation studies across five backbone models (T5-base, Gemma-2B/7B-It, LLaMA 3-8B, LLaMA 3.1-8B), comparing five fusion strategies: Vanilla VQ-VAE, Body+Hand Concat, Body-First, Alternating, and our proposed Hand-First PA-VQVAE. As shown in Figure 9, results consistently demonstrate the superiority of the Hand-First design across all backbones and metrics, particularly in semantic alignment (R-Precision, MultiModal Distance), temporal smoothness (FID), and motion diversity. While Body+Hand Concat improves over the vanilla baseline by decoupling representations, hierarchical variants further enhance performance, with Hand-First achieving the best results (e.g., on LLaMA 3.1-8B, Top-1: 0.398, FID: 0.619, MultiModality: 2.821). This ordering effect is theoretically grounded: hand gestures often encode semantic intent that drives subsequent body dynamics. Modeling hands first provides a natural causal prior, yielding more coherent and expressive motion generation. These results collectively validate that the performance gains stem not merely from hierarchical modeling but specifically from the hand-first fusion order.

VI. CONCLUSION

In this paper, we introduce the MotionGPT-2, a versatile Large Motion-Language Model (LMLM), which can generate and comprehend human motions with general world knowledge of LLMs. Notably, MotionGPT-2 unifies motion-related tasks with multi-modal control signals (e.g., text and single-frame poses) as input by discretizing pose conditions and creating a unified set of instructions from combined textual and pose prompts. By constructing a unified motion-language vocabulary of LLM, we empower the pre-trained LLMs with the ability to integrate the understanding and generation of body kinetics. With well-designed Part-Aware VQVAE, MotionGPT-2 also

demonstrates its versatility in addressing the complex 3D whole-body motion generation task, establishing a strong benchmark for researchers. We envision that MotionGPT-2 paves the way for more practical and versatile motion generation systems, offering a fresh perspective in the field.

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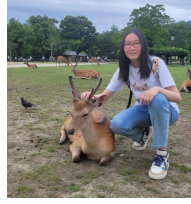
VII. BIOGRAPHY SECTION



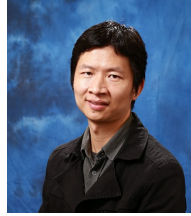
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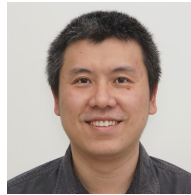
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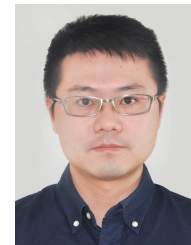
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